

CHEM 101 Topic 11: Acids and Bases — Answer Key

Section 1: Defining Acids and Bases

Problem 1: Arrhenius base. Reasoning: NaOH dissociates in water to produce OH⁻ ions.

Problem 2: Acid: H₂O (donates proton to NH₃); Base: NH₃ (accepts proton from H₂O).

Problem 3: Arrhenius acids always donate protons in water, fitting the BL definition. However, some BL acids donate protons in non-aqueous environments, failing the Arrhenius definition.

Section 2: Water, Ionization, and pH

Problem 4: Basic (or Alkaline). Reasoning: pH > 7.

Problem 5: pH approx 2.60. Reasoning: pH = -log(2.5 x 10⁻³).

Problem 6: 100 times more concentrated. Reasoning: Difference is 2 pH units; 10² = 100.

Section 3: Electrolytes & Net Ionic Equations

Problem 7: HNO₃: Strong electrolyte; HF: Weak electrolyte.

Problem 8: K⁺ and Cl⁻. Reasoning: They remain ions on both sides of the equation.

Problem 9: Ba²⁺(aq) + SO₄²⁻(aq) → BaSO₄(s).

Section 4: Clinical Applications

Problem 10: Acidosis. Reasoning: 7.30 is below the normal range of 7.35 - 7.45.

Problem 11: The base in the antacid reacts with HCl in a neutralization reaction, producing water and a salt, which reduces the concentration of H⁺ ions.

Problem 12: As a strong electrolyte, NaHCO₃ dissociates completely, providing HCO₃⁻ ions that neutralize excess H⁺ in the blood, thereby increasing the pH.